

Program Pembangunan  
Olimpiad Matematik (Siri 1)

**Geometry**

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# 1 Congruency and Similarity of Triangles

## 1.1 Theory

- Two triangles are said to be congruent when their corresponding sides and corresponding areas are equal.
- Two triangles are said to be similar when their corresponding angles are equal and their corresponding sides are proportional.

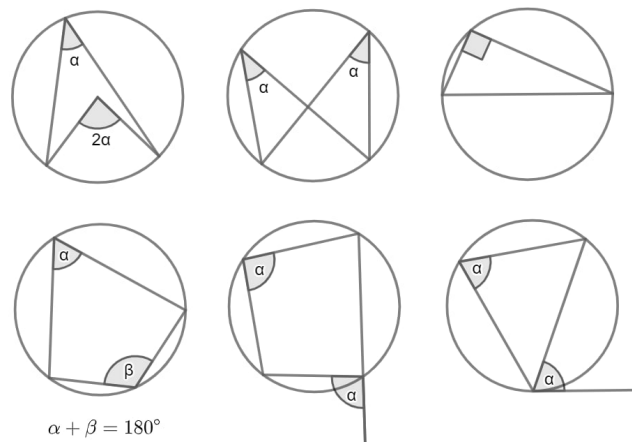
## 1.2 Problems

1. In  $\triangle ABC$ , let  $M$  and  $N$  be the midpoint of  $AB$  and  $AC$  respectively. Prove that  $2MN = BC$ .
2. Prove the Pythagorean Theorem.
3. Prove the Angle Bisector Theorem.
4. In  $\triangle ABC$ , let  $D$  be the point on  $BC$  such that  $AD$  is an angle bisector. Let  $E$  lie on  $AD$  such that  $BE \perp AD$ , and let  $F$  lie on  $AD$  such that  $CF \perp AD$ . Prove that  $AE \times DF = AF \times DE$ .

## 2 Angle Chasing

### 2.1 Theory

- Basic Angle Properties
- Triangle Centers -  $O, I, H, G$
- Angles in Circles



### 2.2 Commonly Used Notation

- $(ABC)$  - Circumcircle of  $\triangle ABC$
- $AB \parallel CD$  -  $AB$  is parallel to  $CD$
- $AB \perp CD$  -  $AB$  is perpendicular to  $CD$
- $AB \cap CD$  - Intersection of  $AB$  and  $CD$

### 2.3 Problems

1. Let  $ABC$  be a triangle and let  $M$  be a point on the ray  $AB$  beyond  $B$  such that  $BM = BC$ . Prove that  $MC$  is parallel to the angle bisector of  $\angle ABC$ .
2. Prove all rectangles are cyclic.

3. Construct acute triangle  $ABC$  with orthocenter  $H$  and feet of altitudes  $D, E, F$ . Find 6 cyclic quadrilaterals in the figure.
4. In triangle  $ABC$ , let the angle bisector of  $\angle BAC$  intersect  $(ABC)$  at  $D$ . Prove that the lengths  $BD = DC$ . (In other words,  $D$  is the midpoint of arc  $BC$  not containing  $A$ .)
5.  $\triangle ABC$  is an isosceles triangle with  $AB = BC$ . Let line  $\ell$  be the tangent to  $(ABC)$  at point  $B$ . Prove that  $\ell \parallel AC$ .
6. Given a cyclic quadrilateral  $ABCD$  where  $AB \parallel CD$ , prove that the lengths  $AD = BC$ .
7. If  $H$  is the orthocenter of  $\triangle ABC$ , prove that the reflection of  $H$  across  $BC$  lies on  $(ABC)$ .
8. If  $I$  is the incenter of  $\triangle ABC$ , prove that  $\angle BIC = 90^\circ + \frac{1}{2} \angle BAC$ .
9. In cyclic quadrilateral  $ABCD$ , let  $I_1$  and  $I_2$  be the incenters of  $\triangle ABC$  and  $\triangle DBC$  respectively. Prove that  $I_1 I_2 BC$  is cyclic.
10. Let  $P$  be a point on the side  $BC$  of  $\triangle ABC$ . The parallel line to  $AC$  through  $P$  intersects the tangent to  $(ABC)$  through  $A$  at a point  $Q$ . Prove  $APBQ$  is cyclic.
11. Let  $ABC$  be an isosceles triangle, such that  $AC = BC$ . Let  $P$  be on the side  $AC$ . The tangent to  $(ABP)$  at point  $P$  intersects  $(BCP)$  at  $D$ . Prove that  $CD \parallel AB$ .
12. Consider a cyclic quadrilateral  $ABCD$ . Let  $M$  be the midpoint of arc  $BC$  not containing  $A$  and  $D$ . Then, let  $AM \cap BC = P$  and  $DM \cap BC = Q$ . Prove that  $APQD$  is cyclic.

### 3 Problems

1. Prove the Power of a Point Theorem.
2. (JBMO SL 2015 G1) Around the triangle  $ABC$  the circle is circumscribed, and at the vertex  $C$  tangent  $t$  to this circle is drawn. The line  $p$ , which is parallel to this tangent intersects the lines  $BC$  and  $AC$  at the points  $D$  and  $E$ , respectively. Prove that the points  $A, B, D, E$  belong to the same circle.
3. (JBMO SL 2012 G2) Let  $ABC$  be an isosceles triangle with  $AB = AC$ . Let also  $\omega$  be a circle of center  $K$  tangent to the line  $AC$  at  $C$  which intersects the segment  $BC$  again at  $H$ . Prove that  $HK \perp AB$ .
4. (JBMO SL 2023 G1) Let  $ABC$  be a triangle with circumcentre  $O$  and circumcircle  $\Omega$ .  $\Gamma$  is the circle passing through  $O, B$  and tangent to  $AB$  at  $B$ . Let  $\Gamma$  intersect  $\Omega$  a second time at  $P \neq B$ . The circle passing through  $P, C$  and tangent to  $AC$  at  $C$  intersects with  $\Gamma$  at  $M$ . Prove that  $|MP| = |MC|$ .